

SIH JUDGE QUESTION & ANSWER DEFENSE DOSSIER

CipherCrest (Sandesh Kavach) — Solution for SIH26159 SecureMailScope:
AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications (NTRO)

Field	Value
Team product	CipherCrest / Sandesh Kavach (v0.7.0-bridge)
Problem ID	SIH26159 — SecureMailScope, NTRO, Software / Blockchain & Cybersecurity
Dossier date	09 Sep 2026 (v3 - adds Day-15 fixes, external corpus, pipeline fix, re-rating)
Sources	Project repo /home/shreyas/projects/CipherCrest + research vault sih26159-securemailscope-ai-crypto-posture + live web verification Sep 2026
Operating principle	Investigation A (claims) vs Investigation B (implementation truth); discrepancies flagged, nothing hidden

HOW TO USE THIS DOSSIER

Numbers in this dossier use the **honest** values only: n_eff 272 (not 500), nested-CV AUC 0.714 (not holdout 0.98), ECOD PR-AUC 0.473 (tooltip-only, never block), 14/20 REAL + 3 info (6 TLS-1.3-opaque greyed). Any slide claiming more will be destroyed. Rehearse every P0 aloud in 20–40 seconds.

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1. Project Identity & SIH Problem Statement

ID: SIH26159 **Title:** SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications. **Organisation:** National Technical Research Organisation (NTRO). **Category:** Software, Theme Blockchain & Cybersecurity. Source mirror: sih2026.vuce.in/ps/SIH26159 (cross-confirmed against SIH 2026 catalogue; treat exact PS PDF as authoritative on finale day).

Background gist: email remains critical for government/enterprise/finance/academia; SMTP/IMAP/POP3 deployments suffer obsolete TLS, weak ciphers, insecure STARTTLS, expired/misconfigured certificates and non-compliance, enabling downgrade, MITM and passive interception. Existing packet tools decode traffic but do not auto-evaluate crypto posture or prioritise risk.

Deliverables checklist (jury will tick): passive PCAP analysis; auto-identify SMTP/IMAP/POP3; STARTTLS detection; TCP-stream + TLS-handshake reconstruction; X.509 chain/expiry/pubkey/sig-algo analysis; TLS version/cipher/KEX identification; deprecated/weak detection; forward-secrecy assessment; AI risk scoring + anomaly detection + prioritisation + mitigation; JSON/PDF/HTML reports + interactive dashboard.

Adjacent statement to distinguish: SIH26106 (AICTE) is *email-content* threat detection + geolocation (phishing/spoof/SPF/DKIM/DMARC). SIH26159 is *transport-crypto posture from PCAP* — no overlap. Say this if the jury conflates them.

2. Executive Project Understanding — Claims (A) vs Truth (B)

2.1 What the team claims

- Passive PCAP → email crypto-posture verdict layer (not a Wireshark wrapper).
- Pipeline: reassembler → TLS parser → X.509 validator → 23-check rule oracle → Rule+XGB/ECOD triage → 20/20 posture JSON + dashboard + reports.
- Offline replay primary, reproducible from lab/pcaps + lab/manifest.json without live capture or decryption.
- Honest coverage 14/20 REAL per-version (6 opaque/info greyed via is_tls13_opaque) vs gateway stretch deferred.
- Rule engine 80% (23 checks, 20 scored + 3 info) + lean ML 20% (XGB Platt + ECOD) as calibrated_prob / anomaly_score on FlowVerdict.
- Ports 25/587/587-STARTTLS + 993 implicit (+110/995 stretch); TLS 1.0/1.1 → Critical per RFC 8996; TLS 1.3 honestly opaque; air-gap offline, no body decrypt.

2.2 What is actually true (B) — with file evidence

- **Real end-to-end chain:** lab/reassembler/reassemble.py → analyzer/parse.py + analyzer/jas.py → validator/chain.py + san_check.py → assessment/rules.py (23) + score.py + policy.py → api/pipeline.py + ml_enrich.py → api/app.py → Postgres/SQLite → dashboard/src. Verified live.
- **Rule oracle real:** 23 findings with RFC/CVE cites; scoring Critical25/High15/Medium7/Low3; policy decide() allow/quarantine/block/flag with opaque clamp + low-confidence downgrade. Files: assessment/rules.py, score.py, policy.py.
- **ML wired but weak-label-limited:** FEATURES_8 frozen; XGB hist + Platt cv=2 GroupKFold canonical; ECOD+IF+hybrid; FlyingSquid m=6 triplet CPI never-train-on-weak-labels. Models on disk: models/risk_clf.pkl (178KB), anomaly.pkl / anomaly_honest.pkl (62KB).
- **Honest numbers (quote these):** 500 envs = 85 original + 415 synthesised (lab/scripts/synth_families.py table A–J + jitter) → canonical 132 distinct → n_eff **272** (DEFF 1.836, ICC 0.3, m=3.79; eval/n_eff_report.json). Nested SGKF5x3 AUC **0.714** vs holdout 0.976 (gap 0.262 disclosed; eval/metrics.json). ECOD PR-AUC **0.473** vs JA4-rarity 0.926 vs IF 0.759 vs hybrid 0.84 (eval/anomaly_baselines.json) — ECOD is tooltip-only, never blocks.
- **Simulated / fixture-backed (disclose):** labels rule-derived weak supervision (verbatim; legal n_eff=10 vs operational 272 vs 500 quality target); Censys/Tranco priors are shared/fixtures/*.json samples (chain fields None for 11/28); MX/MTA-STX/DANE via shared/data/*-fixture.json + mockdns (no live DNS); CatBoost returns simulated AUC 0.82±0.02 when wheel absent (assessment/risk_model.py:448-476); TabPFN/ET-BERT docs-only (torch commented out); manifest SHAs/epochs dummy.
- **Deployment real:** 3-stage Dockerfile (node20 + python3.11 + tshark, :8000, nonroot), compose postgres:16-alpine pinned + demo, init-db/01_schema.sql versioned flows/history, turnup/turndown scripts, CI with postgres service + wheelhouse air-gap.

DANGEROUS DISCREPANCIES — NEVER CLAIM

500 distinct real captures (truth: ~10–13 real pcaps + 415 synth) • n_eff 500 (truth: 272) • AP 0.98 production (truth: nested 0.714) • ECOD blocks threats (truth: 0.473 random, advisory only) • 20/20 full certs on TLS 1.3 (truth: 14/20 REAL, 6 opaque greyed) • CatBoost/TabPFN trained (truth: simulated / docs-only). PS_TRACEABILITY n_risk45/n_eff10 is stale vs Day-14 evidence.

ONE-LINE JURY CLOSE

Gateways stop bad mail; CLM manages live certs; CipherCrest proves — from a PCAP alone — whether the mail transport was cryptographically sound, and tells the analyst what to fix first.

3. Actual Implementation Audit

3.1 Stack & data flow

Python 3.11 | FastAPI 0.115 + uvicorn | React18 + Vite5 | Postgres16 (psycopg3) + SQLite fallback
api/flows.db | scrapy 2.7 primary + tshark 4.2 oracle | cryptography 43 Store/PolicyBuilder | xgboost
1.7.6 + sklearn 1.5.0 Platt + pyod 2.0.5 ECOD + FlyingSquid-custom | Pydantic2 extra=forbid | Docker24
3-stage

POST /analyze (1MiB chunks) -> tmpfile -> reassemble (coverage, pre_tls, transcript) + parse (tls/cert, tshark_used) + jas (JA4/rarity) -> manifest cert match -> validate_chain (skip if opaque) -> evaluate (+history triple) -> score -> 8-col/5-col vector -> risk_clf + ECOD -> FlowVerdict hard-fail -> attach_policy -> Postgres upsert + WS -> GET /flows/report/families/metrics/ws -> dashboard (5s poll, ThreatMatrix, HonestyBanner)

3.2 Failure → effect → detection → mitigation → recovery (excerpt)

Failure	Effect	Detection	Mitigation	Recovery
tshark absent	scapy fallback, less robust reassembly	tshark_used=false flag	ship scapy primary + prefs doc	install tshark, replay
TLS 1.3 cert request	opaque nulls, 14/20 only	is_tls13_opaque=true banner	grey Cert tab, never mock	SSLKEYLOGFILE lab only
ECOD random (0.473)	false positives if blocking	anomaly_baselines.json	advisory tooltip only	use rules+risk for block
CatBoost wheel missing	simulated AUC in CI	risk_model.py:448 branch	label simulated clearly	install wheel, retrain
Postgres down	writes lost	health pg_isready	SQLite mirror api/flows.db	replay upsert on reconnect
Chunk >1MiB / >100MB pcap	413 reject	API 413 + logs	client-side chunking	resume upload

4. Architecture Deep Dive

Components: **reassembler** (scapy 5-tuple, seq buffering, coverage_ratio 1.0 clean / 0.897 jitter, pre_tls_buffer_len, STARTTLS transcript discriminator); **parser** (IANA 11-cipher map, GREASE-16 filter, version/KEX/AEAD/early_data/PSK/ECH); **JAS** (JA4/JA4S/ja4_rarity 0..1, never raw hash for risk); **validator** (dual-store PRIVATE_CA + ca-bundle, per-link pubkey verify, SAN RFC 7817, must-staple, expiry/keysizes/sigalg); **rules+score+policy** (23 checks RFC/CVE-cited, sum-cap-100, Critical≥40/High≥25/Medium≥10/Low, opaque clamp, low-conf downgrade); **features+grouping** (FEATURES_8 frozen sha256, opaque None-mirror, canonical-XXX via canonical_map GREASE-filtered); **risk/anomaly/weak-supervision** (4-experiment harness, CPI K=5 outer fold, GRADE low); **API+DB** (880-line app.py, upsert FOR UPDATE version+1 retry3, pg_notify, GIN queries, matviews); **dashboard** (ThreatMatrix 23×3, CoverageTable, HonestyBanner, WS live, hash deep-link). Bottleneck at 100×: Postgres write + XGB/ECOD inference per flow + WS fan-out; scale by read-replica + inference batching + queue (not yet implemented — admit).

5. Research Deep Dive (H1–H5 verdicts)

Hypothesis	Verdict	What it means for the jury
H1: vanilla IsolationForest PR-AUC>0.80	Strongly falsified as stated (25–40%); hybrid IF+AE/EIF salvage 45–60%	Never claim vanilla IF works; ECOD 0.473 proves it. Defence: hybrid consensus + JA4-rarity ablation + contam sweep.
H2: XGB rule-assisted + isotonic, ECE<0.10	Strongly falsified as isotonic (20–35%); reframed Platt/rank-order 45–60%	Isotonic needs n>1000; team uses Platt sigmoid only + ECE CIs + NDCG@10 fallback. Cite Snorkel +5.81% precedent.
H3: honest TLS-1.3-opaque 14/20 vs mock 20/20	Contested→Corroborated 60–75% (normative triple)	Strongest honesty story: RFC 8446/9846 §4.4 + Wireshark + Zeek no-x509.log. Demo Family-6 grey-out live.

Hypothesis	Verdict	What it means for the jury
H4: co-evolved gauge beats waterfall/rule-only on recall	Falsified on clean recall (30–45%); reframed as triage/MTTR on heterogeneous 50–65%	On clean synthetic grep ties; win is lossy/weberblog + NDCG $\geq$ 0.05 + MTTI/MTTR. Don't claim recall win.
H5: dual-corpus custody $\leq$ 2% erosion	40–55% overall; CIC-as-Bennett 20–35% falsified	Keep weberblog/limbo/Censys/Holz; demote CIC to background artefact-filtered. Custody table: synthetic-vs-dual fails 6/7 — disclose.

6. Problem & Domain Analysis

Root cause: opportunistic STARTTLS + legacy TLS/cipher sprawl + cert mismanagement on mail ports (25/587/993/110/995). Current workflow: analyst opens pcap in Wireshark/tshark, manually hunts ServerHello/certs, runs testssl.sh actively, eyeballs compliance — slow, uncalibrated, unprioritised. Pain: no passive posture score, no FS downgrade timeline, no risk ranking, no analyst-ready report. Users: SOC/forensics/IR/enterprise admins (NTRO PS). Stakeholders: CISOs, auditors, CERT-In. Why current tools insufficient: decoders decode but don't verdict; scanners scan live but can't do forensics on a seized pcap; CLM inventories live certs but not transport sessions. Scale of problem (external): India 52.8M phishing attempts (36.6% APJ, Zscaler 2026); CERT-In 29.44 lakh incidents 2025 (PIB); only 30% orgs completed crypto inventory (IBM IBV 2025) — see §9.

7. Market / Existing Solutions & Competitors

Competitor / Alternative	Core capability	Strength	Weakness vs CipherCrest	Threat
Proofpoint / Mimecast / Abnormal / Microsoft (Gartner Leaders)	Email gateway: content filtering, BEC/phishing	91.4% SGW share (Proofpoint 523 + Mimecast 289 MX); mature ops	Content, not transport-crypto posture; no passive PCAP verdict	Medium — jury will name them; position as complementary
IBM Guardium Cryptography Mgr / SandboxAQ AQtive Guard	Crypto discovery + posture + PQC policy	Centralised inventory, remediation	Live inventory, not PCAP forensics; heavyweight/pricey	Medium-High on posture half
Venafi/Keyfactor/ AppViewX/DigiCert TLM	CLM: issuance, renewal, CA-agility	Cert automation	No session reconstruction / STARTTLS timeline	Low-Medium
Wireshark/tshark, Zeek (ssl.log), Suricata (JA3/JA4), testssl.sh, TLS-Scanner	Decode / detect / active-scan primitives	Free, standard, scriptable	Manual; active-only (testssl); no score/AI/report; Zeek needs scripting	Low as products, High as 'why not script it?' attack — answer with rules+calibration+report evidence

Competitor / Alternative	Core capability	Strength	Weakness vs CipherCrest	Threat
it-pcap-triage / Agentic-Network-Forensics (GitHub)	PCAP→HTML triage pipelines	Generic NSM automation	No X.509 chain + FS + mail-risk model	Low

Defence if judge says a competitor solved it: gateways inspect *content*, CLM inventories *live certs*, decoders *decode* — none takes a *seized passive PCAP* and returns a *calibrated crypto-posture verdict* + *downgrade timeline* + *prioritised remediation* offline. CipherCrest fills that gap; it complements, not replaces, gateways/CLM.

8. Novelty & Differentiation Audit

8.1 Actually novel / defensible

- **TLS-1.3-opaque honesty invariant** enforced in schema + UI (`is_tls13_opaque` → nulls + banner + greyed tab) with normative triple citation — rare in student projects; demo-provable.
- **STARTTLS downgrade timeline** (transcript discriminator + `pre_tls_buffer` injection-vs-stripping split + Family-9 low-conf path) — passive single-PCAP stripping P/R has no standard benchmark; team built its own matrix.
- **Evaluated weak-supervision discipline** (FlyingSquid triplet CPI, never-train-on-weak-labels, GRADE-low disclosure, `n_eff` math) — honesty as a feature.
- **Offline air-gap posture appliance** (wheelhouse + fixtures + mockdns + Wi-Fi-off drill) tailored to NTRO/forensics constraints.

8.2 Not novel / different-but-not-novel

- XGB + Platt, ECOD, JA4, Store/PolicyBuilder validation, FastAPI + Postgres + React dashboard — standard components, honestly assembled.
- Combining decoders + rules + ML + dashboard is *architectural composition*, not algorithmic invention — defend as operational/domain innovation, not new ML.

8.3 Novelty attack matrix (excerpt)

Feature	Existing approach	Difference	Why it matters	Evidence
Opaque handling	Mock full certs / ignore 1.3	Invariant nulls + banner + 14/20	Prevents false assurance	schemas.py validator + App.jsx banner + Family-6 demo
Downgrade timeline	grep 250-STARTTLS	Transcript + coverage + low-conf policy	Distinguishes stripping vs injection; avoids false accuse	reassemble.py + rules 15a/15b + Family-9
Risk triage	Rule list only	Platt-calibrated + ECE CIs + NDCG	Rank-order + calibrated prob, not vibes	metrics_honest.json + risk_model.py
Anomaly	Vanilla IF blocks	ECOD advisory + consensus, 0.473 disclosed	No false blocks; honest about failure	anomaly_baselines.json

9. Statistics & Evidence Audit

Claim	Value	Source / date	Defend	Risk
India most-phished APJ	52.8M attempts, 36.6% APJ	Zscaler ThreatLabz 2026 via CXOToday 11 Aug 2026	Cite for motivation only, not efficacy	Low — reputable, dated 2026
CERT-In incidents 2025	29.44 lakh; 1,530 alerts	PIB 23 Jan 2026; MHA LS data	Proves SOC load	Low
Phishing initial access #3	16% (creds 22%, vuln 20%)	Verizon DBIR 2025	Scope the threat model	Low
Only 30% completed crypto inventory	30%	IBM IBV via Guardium launch 7 Oct 2025	Justifies discovery need	Medium — vendor survey
Team n_eff / AP / ECOD	272 / nested 0.714 / ECOD 0.473	eval/n_eff_report.json, metrics.json, anomaly_baselines.json	Quote honest only; show files live	Critical if inflated

Rule: every dynamic/external number above has a dated source. Internal numbers come from eval/ JSONs — open them live. Anything else: answer 'Not established from available evidence; here is how we would validate it.'

10. Technical Question Bank

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
T01 · P0	What exact PS sentence do you address?	We address passive-PCAP crypto-posture verdicting: reconstruct sessions, find STARTTLS transitions, analyse TLS negotiation, validate certs, AI risk/anomaly + report/dashboard.	Deep: map R1–R8 in PS_TRACEABILITY (stale counts — use Day-14 evidence); demo POST /analyze → FlowVerdict → dashboard. Evidence: api/pipeline.py, assessment/rules.py. Conf: HIGH.
T02 · P0	Show where posture is computed.	Score sums Critical25/High15/Med7/Low3 capped at 100; posture = 100 – risk; level thresholds 40/25/10.	Deep: assessment/score.py + rules.py 23 checks; opaque clamp; history-triple STARTTLS logic. Conf: HIGH.
T03 · P0	Was this real or simulated?	Pipeline real; labels weak-supervised; 85 real + 415 synth; fixtures for DNS/Censys; CatBoost-sim + TabPFN-docs-only disclosed.	Deep: list implemented vs mocked (§2.2); open synth_families.py + fixtures live. Conf: HIGH.
T04 · P1	Why scapy primary, tshark oracle?	Scapy ships everywhere offline; tshark (4 prefs) is the accuracy oracle when present; flag tshark_used records which ran.	Deep: docs/TSHARK.md; lab reassembler tests incl. jitter 0.897 coverage. Conf: HIGH.
T05 · P1	What breaks first at 100× load?	Postgres writes + per-flow inference + WS fan-out.	Deep: upsert FOR UPDATE retry3 + matviews today; admit queue/batch/read-replica needed; give 10/1K/100K/1M scaling story. Conf: MEDIUM.

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
T06 · P2	Why not just Wireshark + script?	Decoders don't verdict: no calibrated score, FS timeline, chain policy, or analyst report; we add evaluated rules + calibration + JSON/PDF/HTML.	Deep: side-by-side: tshark output vs /report for Family-9. Conf: HIGH.
T07 · P3	Prove TLS-1.3 handling isn't mocked.	is_tls13_opaque forces nulls by schema validator; UI greys Cert tab + banner; Family-6 demo shows 14/20.	Deep: shared/schemas.py model_validator + App.jsx HonestyBanner + RFC 8446 §4.4 triple. Conf: HIGH.
T08 · P4	Strongest argument against you?	Weak labels + synth-heavy + nested 0.714 + ECOD random — ML triage is advisory, rules carry the verdict.	Deep: say it before they do; pivot to rules-oracle + honesty + roadmap (n_cal≥300, EIF consensus). Conf: HIGH.

11. AI/ML Question Bank

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
M01 · P0	Why AI at all? Why not rules only?	Rules verdict (80%); ML adds rank-order + calibrated prob + anomaly surfacing (20%) — triage, not truth.	Deep: ablation triad rule-only vs +XGB vs +IF; NDCG@10 Δ−0.005 CI; SHAP coherence; rules block, ML advises. Conf: MEDIUM.
M02 · P0	Dataset? Labels? Leakage?	500 envs (85+415 synth), canonical 132, n_eff 272; rule-derived weak labels (GRADE low); GroupKFold canonical + CPI outer fold; D_prior disjoint; PYTHONHASHSEED 0.	Deep: splits.json + LEAKAGE_REPORT T11 34/34; gap/lineage guards; censys never in risk train. Conf: HIGH.
M03 · P0	Your headline metric?	Nested SGKF5x3 AUC 0.714 primary; holdout 0.976 is optimistic (gap 0.262 disclosed); Brier_joint 0.069 vs 0.22; ECE_quant 0.062 (EW bins [94,6,0,0,0] — use quantile + SmoothECE).	Deep: open metrics.json + metrics_honest.json; never average n116 with n100. Conf: HIGH.
M04 · P1	Why Platt, not isotonic?	n<1000: isotonic overfits (ECE≥0.12); Platt sigmoid + 2000-boot CIs; gated-3-bin, not 5.	Deep: arXiv2109.03480 + sklearn docs per research; ece_hi clamp removed honestly. Conf: MEDIUM.
M05 · P1	ECOD 0.473 — why ship it?	We ship it as advisory tooltip only; blocking uses rules+risk. JA4-rarity 0.926 shows signal is in rarity, not raw hash.	Deep: anomaly_baselines.json; hybrid 0.84; consensus + contam sweep roadmap. Conf: HIGH.
M06 · P2	Overfitting / distribution shift?	Duplicates collapse 73.6% (500→132); LOGO132 + Censys/CORP/CPI checks; weberblog real per-port pin; synth-vs-dual ≤2% fails 6/7 — disclosed.	Deep: generalization.py; admit synth-heavy limits external validity. Conf: MEDIUM.

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
M07 · P3	SHAP proves causality?	No — coherence check only; lab-artefact risk admitted; need crypto-vs-flow-stats counter-vector (open Q3).	Deep: don't oversell explainability. Conf: MEDIUM.

12. Data Question Bank

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
D01 · P0	How many samples? Why?	500 envs quality target; honest n_eff 272; legal verbatim n_eff=10 (rule-derived). 30/bin needs n≈200 honest working.	Deep: splits D1 150/D2 100/D3 30/spare 220/D_prior 50; canonical_map 500→132. Conf: HIGH.
D02 · P1	Representativeness?	Mail.lab.local uniformity = augmentation not diversity; weberblog Thunderbird-only; CIC B-Profile not STARTTLS — demoted to background.	Deep: dual-custody table; Holz/SBA/Censys kept. Conf: MEDIUM.
D03 · P2	New cipher/category in prod?	Unknown → low-confidence path: downgrade Critical→High, flag for review; allowlist + retrain hook retrain_on_denoised().	Deep: policy.py low-conf + FEATURES_VERSION pin. Conf: MEDIUM.

13. Security / Privacy / Legal Bank

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
S01 · P0	How would you attack your own system?	Malformed pcap → parser crash; huge pcap DoS; fixture poisoning; JA4 spoof (curl-cffi); model evasion via weak cipher mimicry.	Deep: 1MiB chunks + 413 cap; Pydantic forbid + opaque invariant; ja4_rarity whitelist only; no private key/body decrypt; audit logging. Conf: MEDIUM.
S02 · P1	Secrets / keys / PII?	No private keys, no body decrypt by design; fixtures offline; Postgres JSONB access-controlled; flows_history versioned for audit.	Deep: air-gap wheelhouse; mockdns; retention/deletion policy needed pre-prod. Conf: MEDIUM.
S03 · P1	Regulatory?	CERT-In incident-reporting + IT Act; DPDP Act for any PII in pcaps (minimise, consent, retention, deletion).	Deep: state as roadmap item with legal review; NTRO deployment needs accreditation. Conf: LOW — say 'requires validation'.

14. Scalability / Deployment / Reliability / Edge Cases

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
SC0 1 · P1	10 → 10M users?	10: compose. 1K: pg pool + matviews. 100K: queue + inference batching + read replica. 1M+: sharded ingest + object store for pcaps (not built — admit).	Deep: bottleneck Postgres → inference → WS. Conf: MEDIUM.
DP0 1 · P0	Prototype → production?	docker compose up (turnup.sh); health /health; versioned schema; CI rebuilds dist; air-gap wheelhouse; Wi-Fi-off drill.	Deep: Dockerfile 3-stage + init-db DDL + download_models Releases fallback. Conf: HIGH.
RL0 1 · P1	Worst inputs?	Empty/malformed/max/jittered/duplicate/TLS-1.3-opaque/adversarial-JA4/concurrent uploads.	Deep: is_malformed path; coverage_ratio; opaque clamp; chunking; idempotent upsert. Conf: MEDIUM.
WR 01 · P0	Why not spreadsheet / SaaS / open-source?	Spreadsheet can't reassemble TCP; SaaS can't take seized pcaps offline; decoders don't score/calibrate/report — show side-by-side.	Deep: mandatory simplicity defence. Conf: HIGH.

15. UX / Economics / Impact Bank

UX: user = SOC/forensics analyst; current workflow Wireshark-hunt; ours = upload pcap → ThreatMatrix + timeline + HonestyBanner + /report in <5s replay, <3s dashboard; poor-connectivity = local compose works offline; mistake path = low-conf flag + versioned history + hash deep-link. **Economics:** build = team time; deploy = 1 VM + Postgres + tshark (no GPU, no licence); per-op ≈ CPU-seconds per pcap (measure before claiming); ROI = MTTI/MTTR reduction (cite Elastic 30m→3m / Google 30m→60s as *vendor analogs, not our control*); payback = analyst-hours saved once measured. **Impact:** NTRO/SOC auditability, CERT-In triage, crypto-inventory gap (30% IBM). **Sustainability:** no GPU, pinned images, fixtures over live fetch.

16. Implementation-Authenticity & Team Defense

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
A01 · P0	Which file does X?	Reassembly: lab/reassembler/reassemble.py. JA4: analyzer/jas.py. Chain: validator/chain.py. Rules: assessment/rules.py. Verdict: api/pipeline.py + app.py.	Deep: open each live; state what team wrote vs library (scapy/cryptography/xgboost) vs fixture. Conf: HIGH.

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
A02 · P1	What did you build vs borrow?	Built: reassembler discriminator, 23-check oracle, FEATURES_8 + canonical grouping, CPI weak-supervision harness, honesty invariant, dashboard triage. Borrowed: scapy, cryptography Store/PolicyBuilder, XGB/ECOD, FastAPI/PG/React.	Deep: never claim primitives as novelty. Conf: HIGH.
A03 · P2	Hardest bug? What failed?	Jitter reassembly 0.897 coverage; TLS-1.3 cert over-claim (fixed via opaque invariant); EW empty bins (fixed via quantile + SmoothECE); ensemble-theatre 0.98 deleted.	Deep: honest failure stories beat polish. Conf: MEDIUM.

17. Cross-Examination Scenarios

X1 — Novelty grind. JUDGE: Isn't this Wireshark + rules? TEAM: Decoders decode; we verdict with evaluated calibration + timeline + report — here is tshark output vs our /report for Family-9. JUDGE: Any undergrad can write rules. TEAM: 23 RFC/CVE-cited checks + history-triple + low-conf policy + GroupKFold/CPI evaluation — open LEAKAGE_REPORT. JUDGE: ML adds nothing (NDCG -0.005). TEAM: Agreed for rank today — ML is advisory; rules block; roadmap re-measures with $n_{cal} \geq 300$. *Lesson: concede fast, pivot to evaluated honesty.*

X2 — Data grind. JUDGE: 500 pcaps real? TEAM: No — 85 + 415 synth, canonical 132, n_{eff} 272; synth generator open. JUDGE: So results meaningless? TEAM: Rules verified on real + synth + weberblog pin; ML numbers labelled GRADE-low weak-supervision; nested 0.714 is the honest headline. JUDGE: Why should NTRO trust synth? TEAM: They shouldn't blindly — dual-custody table shows erosion; deployment needs field pcaps under accreditation. *Lesson: never defend 500-real.*

X3 — Security grind. JUDGE: JA4 spoofable, ECOD random — bypass trivial? TEAM: Yes — which is why raw JA4 never enters risk (rarity only, whitelist) and ECOD never blocks. JUDGE: Then anomaly is theatre. TEAM: Advisory surfacing with disclosed 0.473; consensus + sweep is roadmap; blocking path is rules+risk. *Lesson: show the guardrails, not the score.*

18. Demo Defense + Failure Recovery Scripts

Moment	Judge probe	Calm spoken recovery
Before	What will you show?	Offline replay: upload Family-9 pcap → downgrade timeline + 14/20 banner + /report. Watch the transcript discriminator and low-conf flag.
During	Why is cert tab grey?	That's TLS 1.3 honesty: Certificate is encrypted so we force nulls per RFC 8446 §4.4 — mocking it would be lying. 14/20 REAL.
During	Feed my weird pcap?	Yes — if malformed we show <code>is_malformed</code> + coverage; if unknown cipher, low-confidence flag, never a confident wrong verdict.
After	Real or simulated?	Pipeline real; labels weak-supervised; DNS/Censys fixtures; numbers honest (272 / 0.714 / 0.473). Full map in §2.2.

Moment	Judge probe	Calm spoken recovery
Failure	Network/API/DB down	Air-gap build runs offline; SQLite mirror api/flows.db keeps verdicts; Postgres replays on reconnect.
Failure	Slow / wrong output	Show coverage + versioned history + hash deep-link; quote ECE CIs; file issue with manifest hash.
Failure	Empty dataset	Seed fixtures load via lifespan seed_all; dashboard CoverageTable shows what's missing.

19. Rapid-Fire Round

- Problem? — Mail transport crypto fails silently (legacy TLS, STARTTLS strip, bad certs); analysts hand-decode pcaps.
- Solution? — Passive pcap → evaluated posture verdict + timeline + report, offline.
- Why now? — TLS-1.3 opacity + crypto-agility/PQC pressure + 29.44L CERT-In incidents (2025).
- Innovative? — Opaque-honesty invariant + downgrade timeline + evaluated weak-supervision, not new ML.
- Benefits? — SOC/forensics/IR get prioritised fix-first verdicts in seconds.
- Biggest challenge? — No labelled SMTP-TLS corpus; synth-heavy honesty (n_eff 272).
- Strongest result? — Evaluated rules oracle + honesty + nested/CPI/LOGO discipline (not a big AP).
- Biggest limitation? — ECOD 0.473 advisory-only; synth-heavy; fixtures for DNS/Censys.
- Competitor? — Gateways (content) + Guardium/AQtive (live inventory); we do passive PCAP forensics.
- Why not existing? — Nothing returns calibrated posture + timeline + report from a seized pcap offline.
- Scale? — Compose→queue/batch/replica story; 100× bottleneck admitted.
- Cost? — 1 VM + PG, no GPU/licence; per-op TBD by measurement.
- Secure? — No keys/body decrypt; Pydantic forbid; audit history; JA4-spoof/ECOD guards.
- Deployable? — Yes offline via turnup.sh + Wi-Fi-off drill; accreditation pending.
- Key metric? — Nested AUC 0.714 + Brier/ECE with CIs (not holdout 0.98).
- If it fails? — Low-conf flag + versioned history, never confident-wrong.
- Next? — n_cal≥300 re-measure, EIF consensus, field pcaps, per-port 20×5 matrix (D1–D4).
- Why select? — Only honest passive PCAP→posture+AI-triage built for NTRO forensics reality.

20. Top 20 Judge Attacks (condensed — full bank above)

# At tack	Why dangerous	Correct answer	Evidence	Fix
1. 500-real?	Destroys credibility	'500 captures' → reply '85+415 synth, canonical 132, n_eff 272; generator open'	eval/canonical_map.json + synth_families.py	Rename slides to '500 envs (85 real + 415 synth)'
2. AP 0.98 prod?	Theatre	'Holdout optimistic; nested 0.714 primary, gap 0.262'	metrics.json nested vs holdout	Headline nested only
3. ECOD blocks?	Random blocker	'0.473 advisory-only; rules block'	anomaly_baselines.json	Tooltip-label ECOD

# At tack	Why dangerous	Correct answer	Evidence	Fix
4. 20/20 on 1.3?	False assurance	'14/20 REAL; opaque invariant + banner'	schemas.py + demo	Keep banner always
5. Just Wire shark?	No novelty	Show verdict vs decode side-by-side	pipeline.py + /report	One-slide diff
6. ML adds what ?	NDCG -0.005	'Triage advisory; rules carry verdict'	ndcg_honest.json	ML-advisory labels
7. Is isotonic EC E?	n<1000 fail	'Platt-only + CIs + gated-3-bin'	risk_model.py	Remove isotonic mentions
8. JA4 spoof?	Evasion	'Rarity-whitelist only, never raw hash'	ja4_rarity.py	Keep whitelist
9. Censys live?	Air-gap lie	'Offline fixtures, source:fixture badge'	shared/fixtures	Badge fixtures
10. Cat Boost win?	Simulated	'Simulated when wheel absent — labelled'	risk_model.py:448	Install wheel or drop claim
11. Strip ping proven?	Single-pcap over-claim	'Possible/low-conf + injection split'	rules 15a/15b	Soften to possible
12. Cost /scale?	Ops naivety	'1 VM offline; 100× needs queue — admitted'	compose + roadmap	Add scaling slide
13. PII/keys?	Privacy	'No keys/body decrypt; retention TBD'	pipeline + policy	Write retention note
14. DAN E/M TA-STS ?	Fixture theatre	'INFO-only fixtures, 0 egress'	shared/data	Label INFO-only

# At tack	Why dangerous	Correct answer	Evidence	Fix
15. PQC ready?	Buzzword	'Stretch D5 buffer-gated, not pipeline'	follow-ups D5	Move PQC to roadmap
16. SHAP causal?	Overclaim	'Coherence only; counter-vector open'	open Q3	Soften SHAP slide
17. CIC Bennett?	Dataset misuse	'Demoted to background artefact-filtered'	iteration-001	Remove CIC-as-Bennett
18. 34/34 attacks?	No artefact	'Provenance strings only — needs log'	LEAKAGE_REPORT	Produce log or drop
19. TabPFN/ISO TA?	Docs-only	'Rejected (!torch); docs-only'	requirements	Label docs-only
20. Who maintains?	Bus factor	'Pinned FEATURES_VERSION + history + CI determinism'	coldstorage + CI	Add runbook

21. Claim Audit

Claim	Verdict	Action
Offline replay from lab/pcaps + manifest	SUPPORTED	Keep; demo Wi-Fi-off
23 checks, 20 scored + 3 info; Critical25/High15/Med7/Low3	SUPPORTED	Keep; open rules.py/score.py
14/20 REAL + opaque banner (TLS 1.3)	SUPPORTED	Keep; demo Family-6
500 distinct real captures	MISLEADING	Replace with '500 envs: 85 real + 415 synth; canonical 132; n_eff 272'
n_eff 500 production	PARTIALLY SUPPORTED (quality target)	Headline 272 honest; 500 as target
AP 0.98 / AUC 0.97 production	MISLEADING without qualifier	Headline nested 0.714; holdout labelled optimistic

Claim	Verdict	Action
ECOD anomaly blocks threats	UNSUPPORTED	Re-label advisory-only (0.473)
CatBoost AUC 0.82 trained	REQUIRES VALIDATION (simulated path)	Label simulated or retrain with wheel
TabPFN/ET-BERT benchmarked	MISLEADING	Label docs-only / rejected (!torch)
20/20 full certs incl. TLS 1.3	MISLEADING	Keep 14/20 REAL wording
MTA-STX/DANE/PQC in pipeline	PARTIALLY SUPPORTED (fixtures/stretch)	Move to INFO/roadmap
34/34 hyperplan cross-attacks	REQUIRES VALIDATION	Produce artefact log or drop

22. Weaknesses + Recommended Fixes

Technical: synth-heavy (85→500 via table A–J) — fix D3 per-port 20×5 + field pcaps under accreditation. **Data:** weak labels GRADE-low — fix D4 $n_{cal} \geq 300$ + EIF consensus + LF matrix. **Performance:** holdout/nested gap 0.262; EW bins [94,6,0,0,0] — fix quantile + SmoothECE + gated-3-bin (done, keep). **UX:** Vite bloat risk — code-split or Streamlit fallback. **Security:** JA4-spoof + fixture-poisoning — rarity-whitelist + source:fixture badges + signed fixtures. **Economic:** per-op cost unmeasured — benchmark CPU-s/pcap before finale. **Top-3 pre-judging changes:** (1) rename every '500/ n_{eff} /AP' slide to honest values; (2) D1 opaque-armour + Wi-Fi-off drill green; (3) ECOD tooltip-only + CatBoost-sim labels + PQC-to-roadmap.

23. Final Jury Simulation + Verdict

Hostile drill (excerpt): J: 'Why isn't this testssl.sh on a pcap?' A: 'testssl is active-scan; we do passive forensics with transcript + chain + calibration — different evidence.' J: 'Prove stripping from one pcap.' A: 'We claim possible/low-conf with injection-vs-strip split, not proven — timeline + coverage shown.' J: 'Your ML is worse than rules.' A: 'Agreed today — ML advises, rules decide; NDCG disclosed; re-measure gated.' J: 'Scale to a CERT-In feed?' A: 'Single-node offline today; feed needs queue/shard — not built; here's the bottleneck math.' J: 'What fails completely?' A: 'All-synth training without field pcaps + treating ECOD as blocker — both guarded by disclosure + policy.'

FINAL JURY VERDICT

Strongest: evaluated honesty (opaque invariant + weak-supervision disclosure + nested/CPI discipline) + real pipeline + offline fit for NTRO. **Weakest:** synth-heavy ML triage (nested 0.714, ECOD 0.473) — advisory only. **Most defensible:** 14/20 REAL + downgrade timeline. **Least defensible:** any 500-real / 0.98-prod / ECOD-blocks slide. **Likeliest attack:** 'just Wireshark + rules?' **Most dangerous unanswered:** field-pcap generalisation without accreditation data. **Risks:** technical = synth generalisation; novelty = composition not algorithm; deployment = scale-out unbuilt; credibility = number inflation (fixable in a day). **Three changes:** honest-number slides; Wi-Fi-off + Family-6/9 live drill; ECOD/CatBoost/PQC labels corrected.

24. Source / Evidence Index

ID	Title / location	Type	Supports
INT-PROJ	/home/shreyas/projects/CipherCrest (README, PS_TRACEABILITY, EVIDENCE_Day14, LEAKAGE_REPORT, analyzer/, assessment/, api/, eval/, models/, docs/)	Internal project	Implementation truth + honest metrics

ID	Title / location	Type	Supports
INT-RES	.omnilearn/research/sih26159-securemailscope-ai-crypto-posture (INTENT_SPEC, H1–H5, findings, iteration-001/002, D1–D5)	Internal research	Hypotheses verdicts + gates + roadmap
EXT-SIH	sih2026.vuce.in/ps/SIH26159 (mirror; verify vs sih.gov.in finale PDF)	External/gov	PS text + NTRO + deliverables
EXT-ZSC	Zscaler ThreatLabz 2026 via CXOToday 11 Aug 2026	External/industry	52.8M India phishing
EXT-PIB	PIB 23 Jan 2026; MHA LS 5065	External/gov	CERT-In 29.44L incidents
EXT-DBIR	Verizon DBIR 2025 (Optery/Abnormal summaries)	External/industry	Phishing 16% initial access
EXT-IBM	IBM Guardium Cryptography Manager launch 7 Oct 2025	External/vendor	30% crypto-inventory gap
EXT-NIST	NIST FIPS 203/204/205 (Aug 2024); SP 800-52r2; IR 8547	Standards	PQC + TLS config
EXT-RFC	RFC 8446/9846 §4.4, 5280§6, 3207/2595/8314, 8996, 8461/8460, 7817	Standards	Opaque + STARTTLS + chain rules
INF	Inferences (scaling bottlenecks, cost model)	Inference	Labelled as inference, not fact
ASM	Assumptions (finale PS PDF matches mirror; wheel absent → sim path)	Assumption	Validate on judging day

No-hallucination pledge: statistics above are dated and sourced; internal numbers trace to eval/* .json + code paths named; anything else is marked inference/assumption or 'Not established from available evidence.' Speaking style for judges: 'Yes. We chose X because... We tested X against Y and observed... Our limitation is... We have not validated that yet, so we would not claim...'

25. Fresh Evidence Update — 09 Sep 2026

Eval re-run against latest code (clean venv py3.12, wheelhouse + PyPI fallback, PYTHONHASHSEED=0): eval/tests 35/35 PASS; assessment + shared + analyzer + validator + lab-reassembler 330 passed, 3 skipped; honest-metrics hard gate load_and_validate_honest() PASS. Post-Day-14 commits (stripping ladder, timeline, catalog, coherence) broke nothing. API/DB tests need Postgres and run in CI only.

Suite	Result	Note
eval/tests	35/35 PASS	1 initial failure was missing fastapi in throwaway venv, not repo code
assessment/shared/analyzer/validator/reassembler	330 passed, 3 skipped	80s run; only deprecation warnings
Honest gate (schemas_eval)	PASS	AP 0.976 holdout / nested 0.714 / ECOD 0.473 / n_eff 272 all reconfirmed

25.1 Post-Day-14 delta (what Day-14 evidence does NOT cover)

- **Stripping ladder (8824520)**: mode none emits only check-14; stripping needs lane-A signal or stripped mode; Critical only on 2-prior-upgraded-then-clear-text triple via SQLite last-10 history in api/pipeline.py.
- **UpgradeTimeline.jsx (new, 410 lines)** + RunHistoryTimeline + acceptance/rejection dossier in Reports.jsx — best demo assets, absent from Day-14.
- **Family catalog (179f718, a9bbd60)**: full catalog loads, only family-01..10 pre-seeded demo baseline, rest Not Run standby; pagination-snapping fixed.
- **Live coherence + distinct guard (58453ac)**: live simulated packets cryptographically coherent; Postgres upsert WHERE IS DISTINCT FROM — no score-reset on re-run.
- **Reassembler fix (2a51a17)**: pre_tls_buffer uses last banner before ClientHello — injection-vs-stripping split now correct.
- **Unmerged draft**: .omo/notepads/hyperplan-ciphercrest-ml-remediation (580 envs, n_eff 319 claims) never landed in eval JSONs — never quote to judges.

25.2 Real-protocol pcap (non-synthetic)

File: lab/pcaps/real-mail-starttls-587.pcap (3996 B, 28 packets) + lab/pcaps/REAL-MAIL-PROVENANCE.md + generator /tmp/real_mail_capture.py. Real SMTP dialogue over a real socket (Postfix-faithful banner, 250-STARTTLS, 220 Ready); TLS bytes from a live OpenSSL handshake (TLS 1.2, ECDHE-RSA-AES256-GCM-SHA384, real DER of lab/certs/rsa2048.crt); post-handshake ciphertext is real AES-GCM. Only L2-L4 framing reconstructed (loopback capture kernel-blocked: dumpcap permission denied). tshark dissects it independently. Pipeline verdict: smtp/upgrade, risk 13 Medium, posture 87. Known gap logged: chain-validation finding did not fire on this flow — do not claim chain-validated on it.

25.3 Stale-claim fix status

Location	Stale claim	Required fix	Status
PS_TRACEABILITY.md header + last row	n_risk45/n_eff10, 45 envs, Day-12 evidence	Rewrite to 500 envs / canonical 132 / n_eff 272 + Day-14/15	OPEN
README.md charter + evidence links	n_eff 500 as fact; AP 0.976 bare; ensemble 0.980	Add measured-vs-target line; qualify AP; delete 0.980	OPEN
EVIDENCE_Day14.md headline	HONEST SUCCESS 500 proper distinct	Supersede via EVIDENCE_Day15.md (outline below)	OPEN
App.jsx / Families.jsx / Reports.jsx (16a/17/16b)	MTA-STDS/DANE scored + ACTIVE/pass:true	source:fixture badges; INFO-only; pass:null	OPEN
risk_model.py:448-476 + slides	CatBoost 0.82 bare	Label simulated-when-wheel-absent everywhere	OPEN
Slides/docs	TabPFN/ET-BERT benchmarked; 34/34 log	Label docs-only; produce log or downgrade	OPEN

EVIDENCE_Day15 OUTLINE (write next)

1. Ladder rule diff + triple-history replay log.
2. UpgradeTimeline screenshots + Wi-Fi-off drill log.
3. Catalog: 10 seeded + Not Run standby screenshot.
4. Real-pcap replay log (posture 87).
5. Re-run metrics (35/35 + 330).
6. Stale-claim closure checklist above, each ticked with commit hash.

26. Complete Research Question Bank

Every falsifiable question from the research vault, with the jury-ready answer and the current verdict. Sources: 02-hypotheses/hypothesis-registry.md, 04-synthesis/open-questions.md, 01-background/contradictions-map.md, 05-iterations/iteration-001.md.

26.1 H1-H5 — hypothesis cross-examination

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
H1 · P0	Does IsolationForest anomaly detection actually separate weak/stripped mail-TLS from normal?	No — not vanilla. ECOD PR-AUC 0.473 (near-random) vs JA4-rarity 0.926 on the same split. Anomaly is advisory tooltip only; blocking uses rules plus risk.	Deep: falsification met — vanilla IF strongly falsified as stated (CIC outlieriness 0.17-0.55 precedent; Wang AE gap 30pc F1/111pc AUC). Salvage path is hybrid IF+AE/EIF consensus (Q1 experiment open). Evidence: eval/anomaly_baselines.json; research H1 (02-hypotheses/H1-isolationforest-handshake-pr-auc-co ntest.md). Conf: HIGH.
H2 · P0	Is your XGB just memorizing your own rules (circular)?	Partly — labels are rule-derived, GRADE-low, and we say so. Its value is rank-order plus calibrated probability, judged by NDCG and ECE with CIs, not by F1 alone.	Deep: isotonic at n under 1000 is backwards (needs 1000+, predicts ECE over 0.12) so Platt sigmoid only + 2000-boot CIs + gated-3-bin. Ablation triad rule-only vs +XGB vs +IF required; SHAP coherence gate (2 of cipher/keylen/expiry/FS/chain) open — Q2/Q3. Evidence: metrics_honest.json; H2 file. Conf: MEDIUM.
H3 · P0	Why should we reward 14/20 instead of punishing incompleteness?	Because 20/20 on TLS 1.3 from a passive pcap is cryptographically impossible (RFC 8446/9846 s4.4; Wireshark needs SSLKEYLOGFILE; Zeek emits no x509.log). Anyone showing full certs is fabricating.	Deep: normative triple, no counterevidence found; contested only on scoring psychology (completeness bias, 30pc half-built=zero). Family-6 gate: opaque forces nulls by schema validator. A/B honesty experiment open (Q4). Evidence: shared/schemas.py validator + App.jsx banner + H3 file. Conf: HIGH.
H4 · P1	Does your co-evolved dashboard beat a last-week mock dashboard?	On clean synthetic recall: no — deterministic grep ties (concede). Win is heterogeneous triage: lossy/weberblog parity + NDCG@10 >= 0.05 + MTTI/MTTR, measured not vendor-claimed.	Deep: as-recall strongly falsified; reframed as triage/MTTR 50-65pc. Lossy experiment (tc netem + fragmented Certificate) open — Q5. Evidence: 04-synthesis/findings.md; H4 file. Conf: MEDIUM.
H5 · P1	Does dual-corpus (real pcaps) actually help, or is it decoration?	Composition fixed by falsification: keep weberblog/limbo/Censys/Holz; CIC-as-Bennett demoted to artefact-filtered background (B-Profile is not STARTTLS mail). Custody gate synthetic-vs-dual <= 2pc fails 6/7 — disclosed.	Deep: real pcap added 09 Sep (la b/pcaps/real-mail-starttls-587.pcap , posture 87) is the first dual-corpus repayment. Likert + custody experiment open — Q6. Evidence: REAL-MAIL-PROVENANCE.md; H5 file. Conf: MEDIUM.

26.2 Q1-Q8 — open research questions with current answers

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
Q1 · P2	Hybrid IF+AE: does it reach PR-AUC over 0.80 with CIs non-overlapping vanilla?	Unknown — experiment open. Required: same corpus/vector/split, contam sweep 0.01-0.20, JA4 ablation, weberblog outlier-rate table. Today: quote 0.473 honest, hybrid 0.84 non-gated. Do not promise 0.80.	
Q2 · P2	Platt vs isotonic at n approx 80: does Platt reach ECE under 0.10?	Partially answered: Platt-only shipped + ECE CIs + gated-3-bin; isotonic rejected per theory. Remaining: head-to-head with non-overlapping CIs or fall back to NDCG triage framing.	
Q3 · P2	SHAP top-3: crypto signals or lab artefacts?	Open — the circularity killer. Required: crypto-vector vs flow-stats counter-vector SHAP comparison. Until run: never claim explainability beyond coherence intent.	
Q4 · P2	Honest-vs-mock A/B: does honesty gain 5+ points?	Open. The finale IS the experiment — run the Family-6 grey-out demo and the Q and A battery (is_tls13_opaque, tshark vs x509.log, ServerHello.random, SSLKEYLOGFILE) as the probe set.	
Q5 · P2	Lossy/weberblog parity: does the reassembler hold F1 over 95pc where grep drops?	Open. Required: netem + fragmented-handshake harness vs tshark oracle + NDCG/MTTR triage measure. Real pcap (09 Sep) is step one; lossy harness is step two.	
Q6 · P2	Dual-corpus custody within 2pc + Likert win?	Open. Real pcap parses and verdicts sanely (87, no overclaim) — first custody datapoint. Remaining: parse-rate table + blind Likert + CIC replacement (Holz/SBA).	
Q7 · P3	Per-port rule: what is Critical/High vs Info on 25/587/993?	Partially answered in code (per-port scoping + check-14 vs stripping split, Sep 7 ladder). Remaining: 20-checks-x-ports compliance matrix validated against Ultimate-PCAP-style real mix.	

ID	QUESTION (P-level)	SPOKEN ANSWER (20-40s)	DEEP DEFENSE + EVIDENCE
Q8 · P3	JA4: signal or pivot-not-proof?	Answered as policy: rarity-whitelist only, never raw hash; JA4-only under 0.60 expected. Remaining: ablation Δ measurement. Jury line: pivot, not proof; spoofable (curl-cffi), volatile, JA4X coverage under 1/3.	

26.3 Contradictions C1-C7 and gaps G1-G6 (map)

ID	Contention	Resolution in this project
C1 IF vs AE	IF 0.80+ vs AE beats IF by 30pc/111pc	Vanilla IF falsified; hybrid path open (Q1); ECOD advisory
C2 XGB gain vs circularity	Calibrated gain vs rule-mimic tautology	Disclosed weak supervision; Platt + NDCG framing; SHAP gate open (Q2/Q3)
C3 opaque honesty vs mock completeness	Triple-normative truth vs completeness bias	Honesty invariant shipped; A/B is the finale itself (Q4)
C4 co-evolved vs waterfall	Gauge-from-Day-1 vs polish-last-week	Recall tie conceded; triage/MTTR reframe; lossy test open (Q5)
C5 rule-only sufficiency	Rules suffice for flagging vs ML triage value	Rule80+ML20: rules decide, ML advises; NDCG disclosed
C6 implicit (RFC8314) vs STARTTLS baseline	Both true per-scope	Per-port scoping + Sep-7 ladder; matrix validation open (Q7)
C7 PQC/ECH depth vs scope	PQC readiness vs passive-scope creep	Stretch D5 buffer-gated; fixtures INFO-only
G1-G6	No labeled SMTP-TLS corpus; no passive stripping P-R; no pcap-to-limbo bench; 0-RTT/stapling gaps; JA4 evaluation gap; per-port matrix gap	PS sanctions synthesis (S00); own matrix built; real pcap Sep 9; 0-RTT/ECH INFO-only; JA4 pivot policy

27. Rating vs Best-Project-for-This-PS

Method: scored against the best project plausibly achievable for SIH26159 (field-validated on real NTRO pcaps, per-port matrix measured, calibrated ML on $n \geq 300$ real labels, scale-out ingest, published posture benchmark). 10 = that ceiling. No inflation.

Dimension	Score/10	Why (one line)
Problem quality	9	NTRO passive-forensics gap is real and well-scoped

Dimension	Score/ 10	Why (one line)
Problem-solution fit	8	Deliverable-for-deliverable PS coverage; live/scale ends unbuilt
Novelty	5	Architectural + honesty + ladder innovation; no algorithmic invention
Technical depth	7	Real pipeline and eval discipline; ML component weak
Implementation quality	7	365 tests green, versioned DB, fixtures/sim seams disclosed
Research quality	8	Falsifiable vault with gates, precedents, and overturn criteria
Evidence quality	6	Honest but synth-heavy, docs stale until Day-15
Feasibility	8	Offline appliance is credibly deployable
Practicability	6	Needs field pcaps + accreditation + scale-out
Scalability	4	Single node; queue/shard admitted unbuilt
Security	6	Threat-modeled, air-gap; no pentest, fixture trust open
UX	7	Analyst-first; timeline + dossier are finale-grade
Sustainability	7	No GPU, pinned images, fixtures over live fetch
Impact	6	SOC triage value argued, MTTR unmeasured
Differentiation	6	Narrow but real USP in crowded adjacency
Deployment readiness	6	Compose works offline; production hardening pending
Judge defensibility	7	Honesty story is elite if numbers stay honest

OVERALL: 6.5 / 10 vs best-for-PS

Strongest-in-class on honesty and eval discipline (few student teams disclose n_{eff} math, nested gaps, and random baselines). Mid-pack on ML efficacy and scale. Path to 8-8.5 before judging: Day-15 evidence + field pcap + per-port matrix measured + n_{cal} re-measure with CIs + per-op cost benchmark. Ceiling risks that cap the score regardless: synth-heavy training, advisory-only anomaly, single-node ingest.

Where it beats the hypothetical best: almost nothing beats it on forensic honesty mechanics today (opaque invariant, ladder, GRADE-low disclosure) — most best-case projects would still mock TLS-1.3 certs. **Where it loses:** field validation, measured per-port compliance, calibrated ML on real labels, throughput story, costed deployment. **Three gap-closers that move the number most:** (1) one real field pcap replayed live (+0.5); (2) per-port 20-check matrix measured on real mix (+0.5); (3) Day-15 closing every stale claim (+0.5).

28. v3 Addendum — Day-15 Fixes, External Corpus, Re-rating (09 Sep 2026)

What changed after v2: (1) EVIDENCE_Day15 written with live ladder/timeline/catalog/real-pcap logs. (2) Six stale-claim fixes applied to README, PS_TRACEABILITY, App.jsx, Families.jsx, Reports.jsx (WEAK SUPERVISION verbatim preserved; README held at 386 lines). (3) Pipeline mode-derivation bug found by the per-port matrix and fixed same day. (4) Two external real captures downloaded, replayed, committed. (5) Full verification: 391 passed, 3 skipped.

28.1 Pipeline fix (found by measurement, not review)

Symptom: family-09 verdict **implicit / Critical-44 with zero STARTTLS findings** — the line-83 fallback treated TLS version unknown as implicit, making the ladder Critical tier unreachable live. Fix (api/pipeline.py:83): unknown/no-handshake is never implicit; cleartext falls to none so check-14 fires. Verified: family-09 none + check-14 High; family-06 implicit Low 94; family-01 and real pcap upgrade 87. Regression: api/tests/test_pipeline_mode.py (3 tests). Open cosmetic note (Day16): cleartext flows also collect KEX/FS Highs.

28.2 External real corpus (measured)

Capture	Provenance	Verdict	Use
Wireshark wiki smtp.pcap (2009, 60 pkts)	Exim 4.69 banner, STARTTLS advertised, AUTH LOGIN; via briliant-ben mirror	smtp/upgrade, risk 37 High, posture 63	Cleartext-era negative; parses, correctly untrusted
Weberblog Ultimate mail slice (256 KB, ports 25/ 110/143/465/587/9 93/995)	Direct pcapng.gz + tshark port slice; mail.webertest.net sessions	smtp/upgrade, risk 50 Critical, posture 50 (TLS1.2 UNKNOWN-c02b)	Multi-port real mix; exposed 11-entry cipher-map gap (Day16)
Local real-protocol (v2 §25.2)	Live OpenSSL handshake + real DER; framing reconstructed	smtp/upgrade, risk 13 Medium, posture 87	Positive control; generator lab/scripts/re al_mail_capture.py

Dual-corpus status: synthetic 500-env + local real-protocol + 2 external real captures, all replayed with sane verdicts and zero crashes. Full source list incl. not-yet-fetched (Lekensteyn TLS+keys via browser, malspam zips, CIC document-only): lab/pcaps/external/PROVENANCE.md.

28.3 Claim-closure status (v2 §25.3 → now)

All six CLOSED in code: traceability rewritten, README measured-vs-target lines, Day-14 superseded by Day-15, MTA-STS/DANE fixture-badged in UI, CatBoost-sim noted, TabPFN/34-34 stance recorded. Evidence: git diff of README.md, PS_TRACEABILITY.md, App.jsx, Families.jsx, Reports.jsx (Sep-09 commit).

REVISED OVERALL: 7.0 / 10 (was 6.5)

Upgrades banked: evidence quality 6→7 (Day-15 + matrix + external corpus), judge defensibility 7→8 (all six stale claims closed in code, ladder live, regression-tested), implementation 7→7.5 (bug found by measurement and fixed same day). Scalability (4), ML efficacy (nested 0.714, ECOD advisory), and field-validation depth are unchanged — they cap further gains. Next +0.5 each: measured per-port matrix formalized with custody table (done in repo, needs Likert), n_cal re-measure, per-op cost benchmark. Path to 8.0 remains as stated in §27.

End of dossier — rehearse P0 aloud, drill Wi-Fi-off, and concede weaknesses before the jury finds them.